



- A train takes 18 s to pass completely through a station 162 m long and 15 s through another station 120 m long, at the same speed. What is then length of the train in metre?
(a)70 (b)80 (c)90 (d)100
- In a row of children facing North, Shamika is third to the right of Nikhil, who is 17th from the right end of the row. Ravi is 5th to the left of Shamika and is 20th from the left end. Totally how many children are there in the row?
(a)37 (b)38 (c)39 (d)None
- Given that
 - Some apples are blackberries.
 - Some doughnuts are apples.
 - No coconut is a doughnut.
 - All blackberries are coconuts.

Which of the following statements is false?

- Some blackberries are doughnut
- Some coconut are apples
- All coconuts are not apples
- All doughnuts are not coconuts

Direction[Q. Nos.54-56]In a family of 6 persons, there are two couples. The lawyer is the head of the family and has two sons –Mukesh and Rakesh both teachers.Mrs. Reena and her mother-in-law both are lawyers. Mukesh's wife is a doctor and they have a son, Ajay.

- Which of the following is definitely a couple?
(a)Lawyer-Teacher (b)Doctor-Lawyer
(c)Teacher-Teacher (d)None of these
- What is the profession of Rakesh's wife?
(a)Teacher (b)Lawyer
(c)Doctor (d)Cannot be determined
- What is/was Ajay's grandfather's occupation?
(a)Teacher (b)Lawyer
(c)Doctor (d)Cannot be determined
- Find the missing element in the series.
A, CD, GHI, ..., UVWXY
(a)LMNO (b)MNOP
(c)NOPQ (d)QPQR

- In a code language, 'FRIEND' is coded as 'GTLIST'. Which of the following is coded as 'HWDVT' in that language?
(a)HAPPY (b)GUARD
(c)BEADS (d)SPEED
- There are four brothers Alan, Bob, Carl and Dave. Dave is two years older than Bob. Bob is one year younger than Carl. Alan, who is 34, is two years younger than Carl. Who is the oldest?
(a)Alan (b)Bob
(c)Carl (d)Dave

Directions[Q. Nos. 60-62]An employee has been assigned the task of allotting offices to six of the staff members. The offices are numbered 1-6. The offices are arranged in a row and they are separated from each other by six foot high dividers. Hence, voices sound and cigarette smoke flow easily from one office to another.

Miss. Robert needs to use the telephone quite often throughout the day. Mr. Mike and Mr. Brown need adjacent offices as they need to consult each other often while working. Miss. Hardy is a senior employee and has to be allotted the office number 5, having the

biggest window. Mr. Donald requires silence in the offices next to his and Mr. Tim prefers to be as away as possible from Miss. Robert. Mr. Donald and Mr. Hardy are all smokers. Miss. Hardy finds tobacco smoke allergic and consecutively the offices next to hers to be occupied by non-smokers. Unless specifically stated all the employees maintain an atmosphere of silence during office hours.

- The ideal candidate to occupy the office farthest from Mr. Brown would be
(a)Miss. Hardy (b)Mr. Mike
(c)Mr. Tim (d)Mr. Donald
- The three employees who are smokers, should be seated in the offices
(a)1, 2 and 4 (b)2, 3 and 6
(c)1, 2 and 6 (d)1, 2 and 3

- The ideal office for Mr. Mike would be
(a)2 (b)6
(c)1 (d)3
- A doctor said to his compounder 'I go to see the patients at their residence after every 3 h 30 min. I have already gone to the patient 1 h 20 min ago and next time I shall go at 1:40 pm. 'At what time this information was given to the compounder by the doctor?
(a)11:30am (b)11:20 am
(c)10:10 am (d)None of these

- Which pair of number comes next in the following series?
42 40 38 35 33 31 28
(a)25 22 (b)26 23
(c)26 24 (d)25 23

Directions [Q. Nos. 65-66]

- All G's are H's
 - All G's are J's or K's
 - All J's are K's are G's
 - All L's are K's
 - All N's are M's
 - No M's are G's
- If no P's are K's, then which of the following must be true?
(a)All P's are J's (b)If any P is a G, it is a J
(c)No P is an H (d)If any P is an H, it is a G
 - Which of the following is inconsistent with one or more of the conditions?
(a)All H's are G's
(b)All H's that are not G's are M's
(c)Some H's are both M's and G's
(d)No M's are H's
 - Shyam is taller than Pradeep and Pradeep is as tall as Anurag. But Anand is shorter than Suresh, who is as tall as Anurag. If Pradeep is taller than Praveen, then who is the tallest of all?
(a)Pradeep (b)Praveen
(c)Suresh (d)Shyam
 - When Rajveer was born his father was 32yr older than his brother and his mother was 25yr older than his sister. If Rajeev's brother is 6 yr older than Rajeev and his mother is 3yr younger than his father, then how old was Rajeev's sister when he was born?
(a) 15yr (c) 7 yr





MAARULA CLASSES

TARGET- NIMCET / CUET.PG By: Amit Katiyar (MCA-JNU)

JAMIA ORIGINAL
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SCAN QR OF OUR APP MAARULA MATHEM

- (b) 14yr (d) 10yr
19. Dhoni starts from his office at 8 am on a Sunday morning, travels 10 km towards 10km towards West and then turns to his left and walks 8 km. Then, he again turns to his left and walks 4 km and then stops. What is the shortest distance to his office from the point, where he stopped?

(a) 18 km (b) 8km (c) 10 km (d) None

20. A treasure chest has less than 100 gold coins. The number of coins is

(a) One more than a multiple of 3
 (b) Two more than a multiple of 4
 (c) Three more than a multiple of 5 and
 (d) Four more than a multiple of 6

How many coins are there in the chest?

(a) 58 (b) 88 (c) 98 (d) 38

21. Read the statements and then decide which of the conclusions logically follow:

Statements

- (i) All mangoes are golden in colour.
 (ii) No golden coloured things are cheap.

Conclusions

- (i) All mangoes are cheap.
 (ii) Golden coloured mangoes are not cheap.
 a) Only conclusion I follows
 b) Only conclusion II follows
 c) Either I or II follows
 d) Neither I nor II follows

- Directions** [Q. Nos. 72-73]: A blacksmith has five iron articles A, B, C, D and E, each having a different weight.

- (a) Weighs twice as much as B.
 (b) Weighs four and half times as much as C.
 (c) Weighs half as much as E.
 (d) Weighs less than A but more than C.

22. Which of the following article is heaviest in weight

(a) A (b) B (c) C (d) D

23. Which of the following represents the descending order of weights of the articles?

(a) A, B, E, D, C (b) A, B, C, D, E
 (c) B, D, E, A, C (d) C, D, E, B, A

- Directions** [Q. nos. 74-76]: There are three switches A, B and C which can be in ON/OFF position. Their settings change as per the following rules:

- (i) If A is the only switch as ON, change B to ON.
 (ii) If A and B are only switches as ON, change C to ON.
 (iii) If all three switches are ON, change C to OFF.
 (iv) For all other situations, all switches in On are changed to OFF and all switches in OFF are changed to ON.

24. If switches A and B are ON and C is OFF, then their changed settings will be

(a) A ON, B OFF, C OFF (b) A ON, B ON, C ON
 (c) A ON, B OFF, C ON (d) A OFF, B ON, C OFF

25. If only B is ON, then the changed setting, then which of the following could have been the original setting?

(a) A ON, B ON, C ON (b) A ON, B ON, C OFF

(c) A ON, B OFF, C ON

(d) A OFF, B OFF, C ON

26. If only B is ON in the changed setting, then which of the following could have been the original setting?

(a) A ON, B OFF, C OFF (b) A ON, B ON, C ON
 (c) A ON, B OFF, C ON (d) A OFF, B OFF, C OFF

27. If the third day of a month falls on Friday, then what day will be on the fourth day after twenty first of the month?

(a) Monday (c) Saturday
 (b) Tuesday (d) Thursday

28. Ana is a girl and has the same number of brothers and sisters. Andrew is a boy and has twice as many sisters as brothers. Ana and Andrew are the children of Emma.

How many children does Emma have?

(a) 2 (c) 5
 (b) 3 (d) 7

- Directions** [Q. Nos. 79-81]

1. Anu is taller than Cini.
 2. Eenu is shorter than Binu.
 3. Anu is shorter than Danny.
 4. Eenu is taller than Anu.

29. The best answer to 'who is the tallest?' is

(a) Dany (c) Dany or Binu
 (b) Binu (d) Both Dany and Binu

30. Who is the shortest?

(a) Cini (b) Anu or Cini
 (c) Eenu (d) Insufficient data to conclude

31. Which of the following statements would help to logically order the persons according to their heights?

(a) Binu is 7 feet tall
 (b) Dany and Binu do not have equal height
 (c) Eenu is the tallest in the group
 (d) Dany is the tallest in the group

32. Karan and Arjun run a 100 m race, where Karan beats Arjun by 10 m. To do a favour of Arjun, Karan starts 10 m behind the starting line in a second 100 m race. They both run at their earlier speeds. Which of the following is true in connection with the second race?

(a) Karan and Arjun reach the finishing line simultaneously
 (b) Karan beats Arjun by 1 m
 (c) Arjun beats Karan by 11 m
 (d) Arjun beats Karan by 1 m

33. In a cricket season, India defeated Australia twice, West Indies defeated India twice. Australia defeated West Indies twice. India defeated New Zealand twice and West Indies defeated New Zealand twice. Which country has lost most number of times?

(a) India (b) Australia
 (c) New Zealand (d) West Indies

34. Pointing to a woman, Nirmal said, "She is the daughter of my wife's grandfather's only child". How is the woman related to Nirmal?

(a) Wife (b) Sister-in-law
 (c) Sister (d) None of these

- Direction** [Q. Nos. 35-37] These are five persons A, B, C, D and E standing on six steps numbered 1, 2, 3, 4, 5 and 6 from the bottom. At most one person is standing on each step. The step number on which





A is standing, is two less than that of C. Step number on which B is standing is one more than that D.

35. If A is standing on step 1, which of the following is true?
(a) B is standing on step 2
(b) C is standing on step 4
(c) E is standing on step 3
(d) D is standing one step higher than C
36. If D is standing on step 1, on which step A could be standing?
(a) 2 or 4 only (b) 3 or 5 only
(c) 3 or 4 only (d) 4 or 5 only
37. If there are two steps in between the steps on which A and D are standing, A must be standing on which of the following steps?
(a) 3 (b) 4 (c) 5 (d) 6
38. From the information given below:
A*B means A and B are of the same age.
A-B means B is younger than A.
A+B means A is younger than B.
What does Sachin * Mohan - Ravi mean?
(a) Sachin is youngest (b) Ravi is youngest
(c) Sachin is oldest (d) Mohan is oldest
39. Jimmy saw the time while going to the tennis court. He saw the hour hand is 20° away from 4. After he returned from tennis court, he noticed that the hour hand is 20° away from 4. If he took ten minutes to go to tennis court and he walked at the same speed while going to the tennis court and while returning, how much time did he spent at the tennis court?
(a) 60 min (b) 80 min
(c) 70 min (d) 50 min
40. There are 8 balls looking alike, seven of which have equal weight and one is slightly heavier. The weighing balance is of unlimited capacity. Using this balance, the minimum number of weighings required to identify the heavier ball is
(a) 1 (b) 2 (c) 3 (d) 4

ANSWER

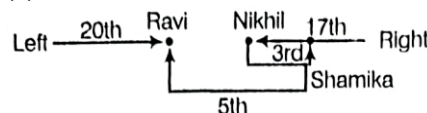
1.	C	2.	B	3.	A	4.	A	5.	C
6.	D	7.	B	8.	B	9.	D	10.	D
11.	D	12.	D	13.	A	14.	C	15.	B
16.	C	17.	D	18.	D	19.	C	20.	A
21.	B	22.	A	23.	A	24.	B	25.	C
26.	B	27.	C	28.	D	29.	C	30.	A
31.	D	32.	D	33.	D	34.	A	35.	D
36.	C	37.	B	38.	B	39.	A	40.	c

SOLUTION

1. (c) Let the length of the train be x m.

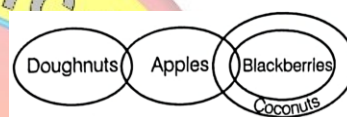
$$\begin{aligned} \text{Then, } \frac{x+162}{18} &= \frac{x+120}{15} \\ \Rightarrow \frac{x+162}{6} &= \frac{x+120}{5} \\ \Rightarrow 6x+720 &= 5x+810 \\ \Rightarrow x &= 90 \text{ m} \end{aligned}$$

2. (b)



- $\therefore$ Ravi's position from the left = 20th
and Ravi's position from the right = $17 + 2 = 19$ th
 $\therefore$ Number of children in the row = $(20 + 19) - 1 = 38$

3. (a) According to the statements,



Hence, some blackberries are doughnuts, is definitely, false.

Solutions [Q. Nos. 54 to 56]

Reena's mother-in-law $\rightarrow$ head of the family $\rightarrow$ Lawyer
Rakesh $\rightarrow$ Married $\rightarrow$ Teacher
Mukesh $\rightarrow$ Married $\rightarrow$ Teacher
Reena $\rightarrow$ Rakesh's wife $\rightarrow$ Lawyer
Mukesh's wife $\rightarrow$ Doctor
Ajay $\rightarrow$ Mukesh's son
Ajay's grandfather $\rightarrow$ Unknown profession

4. (a) Rakesh (Teacher) and Reena (Lawyer) is definitely a couple.
5. (c) Rakesh's wife's profession is Lawyer.
6. (d) We cannot determine the Ajay's grandfather occupation.
7. (b) The pattern of series is as follows:

$$A \xrightarrow{(B)} CD \xrightarrow{(EF)} GHI \xrightarrow{(J,K,L)} MNOP \xrightarrow{(Q,R,S,T)} UVWXY$$

8. (b) As,

$$\begin{aligned} F &\xrightarrow{+1} G & \text{Similarly, } G &\xrightarrow{+1} H \\ R &\xrightarrow{+2} T & U &\xrightarrow{+2} W \\ I &\xrightarrow{+3} L & A &\xrightarrow{+3} D \\ E &\xrightarrow{+4} J & R &\xrightarrow{+4} V \\ N &\xrightarrow{+5} S & D &\xrightarrow{+5} I \\ O &\xrightarrow{+6} J \end{aligned}$$

9. (d) Given, Alan's age = 34 yr
Carl's age = $34 + 2 = 36$ yr
Bob's age = $36 - 1 = 35$ yr
Dave's age = $35 + 2 = 37$ yr
Hence, Dave is the oldest among them.





Solutions (Q. Nos. 10 to 12]

1	2	3	4	5	6

(Mr. Donald) (Mr. Tim) (Mr. Mike) (Mr. Brown) (Miss. Hardy) (Miss. Robert)
Smoker smoker smoker Non-smoker Non-smoker Non-smoker

10. (d) Mr. Donald's office is farthest from Mr. Brown.
11. (d) The three smoker employers of the offices should be seated in 1, 2 and 3.
12. (d) The ideal office for Mr. Mike would be 3.
13. (a) Doctor's previous time to see the patients

$$= 1:40 \text{ pm} - 3 \text{ h } 30 \text{ m} = 10:10 \text{ am}$$

So, the information given to the compounder by the doctor

$$= 10:10 \text{ am} + 1:20 = 11:30 \text{ am}$$

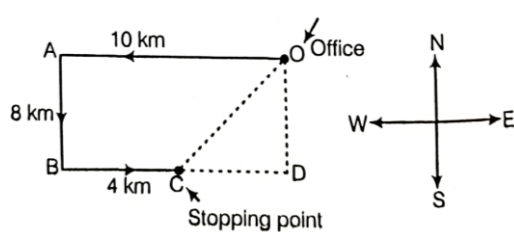
14. (c) Series is increasing in three consecutive terms like
42 40 38; 35 33 31; 28 26 24

Solutions [Q. Nos. 15 to 16]



15. (b) If no P's are K's, then if any P is a G, it is a J, must be true.
16. (c) Some H's are both M's and G's is inconsistent with one or more of the conditions.
17. (d) According to the question,
Shyam > Pradeep = Anurag Suresh > (Anand, Praveen)
Hence, Shyam is tallest among them.
18. (d) Suppose,
Rajeev's age = 0 (Born)
Then, Rajeev's brother's age = 6yr
Rajeev's father's age = 6 + 32 = 38 yr
Rajeev's mother's age = 38 - 3 = 35 yr
Rajeev's sister's age = 35 - 25 = 10 yr

19. (c) Dhoni's walking directions are as follows:

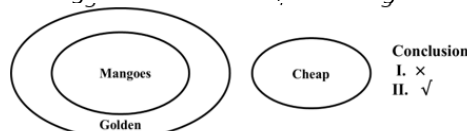


$$\begin{aligned} \therefore OD &= AB = 8 \text{ km} \\ \Rightarrow CD &= BD(AO) - BC \\ &= (10 - 4) \text{ km} \\ &= 6 \text{ km} \end{aligned}$$

$$\begin{aligned} \therefore \text{Required distance (OC)} &= \sqrt{OD^2 + CD^2} \\ &= \sqrt{8^2 + 6^2} = \sqrt{64 + 36} \\ &= \sqrt{100} = 10 \text{ km} \end{aligned}$$

20. (a) By option, number 58 follows the statements rule.
 $583 \times 19 + \boxed{1}$
 $584 \times 14 + \boxed{2}$
 $585 \times 11 + \boxed{3}$
 $586 \times 9 + \boxed{4}$

21. (b) According to the statements, Venn diagram is as follows:



Hence, only conclusion II follows.

Solutions [Q. Nos. 22 to 23]

According to the question.

$$A = 2B \quad \dots(i)$$

$$B = 4.5 C \quad \dots(ii)$$

$$C = \frac{1}{2} D \quad \dots(iii)$$

$$D = \frac{1}{2} E \quad \dots(iv)$$

$$A > E > C \quad \dots(v)$$

Combining all the equations,

$$A > B > E > D > C$$

22. (a) A is the heaviest article in weight.

23. (a) A, B, E, D, C

Solutions [Q. Nos. 24 to 26]

$$A \text{ (only ON)} B \text{ (ON)} \quad \dots(i)$$

$$A \text{ (ON)} + B \text{ (ON)} \rightarrow C \text{ (ON)} \quad \dots(ii)$$

$$A \text{ (ON)} + B \text{ (ON)} + C \text{ (ON)} \rightarrow C \text{ (OFF)} \quad \dots(iii)$$

Except,
ON $\rightarrow$ OFF and OFF $\rightarrow$ ON

24. (b) If switches A and B are ON and C is OFF, their changed settings will be A ON, B ON and C ON.

25. (c) If only B is ON, the changed setting will be A ON, B OFF and C ON.

26. (b) If only B is ON in the changed setting, the original setting could have been A ON, B OFF and C ON.

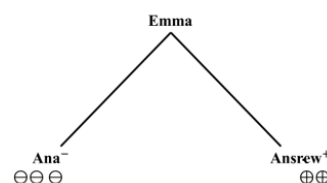
27. (c) $\therefore$ Fourth date after 21st = 21 + 4 = 25
Then,

3 10 17
↑ ↑ ↑

1st Friday 2nd Friday 3rd Friday 4th Friday

$\therefore$ Required day (25th date) = Friday + 1 = Saturday

28. (d)



$$\begin{aligned} \therefore \text{Number of children} &= \ominus + \ominus + \ominus + \text{Ana}^- + \text{Andrew}^+ \\ &= \ominus + \ominus + \ominus + \oplus + \oplus = 7 \end{aligned}$$

Solutions [Q. Nos. 29 to 31]

$$\text{Anu} > \text{Cini} \quad \dots(i)$$

$$\text{Binu} > \text{Eenu} \quad \dots(ii)$$

$$\text{Dany} > \text{Anu} \quad \dots(iii)$$

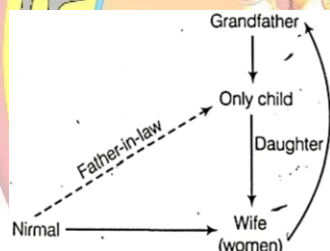
$$\text{Eenu} > \text{Anu} \quad \dots(iv)$$

Combining all the equations,

$$(\text{Dany, Binu}) > \text{Eenu} > \text{Anu} > \text{Cini}$$



29. (c) Dany or Binu is the tallest among them.
30. (a) Cini is the shortest among them.
31. (d) Dany is the tallest in the group, which would help to logically order the persons according to their heights.
32. (d) In 1st 100 m race, distance covered by Karan = 100 m and distance covered by Arjun = 90 m
In 2nd 100 m race, distance covered by Karan = 110 m and distance covered by Arjun = ?
100 m of Karan = 90 m of Arjun
110 m of Karan = $\frac{90 \times 110}{100} = 99$ m of Arjun
∴ Karan beats Arjun by 1 m.
33. (c) India (+2) → Australia (-2)
West Indies (+2) → India (-2)
[+ → win]
[- → loss]
Australia (+2) → West Indies (-2)
India (+2) New Zealand (-2)
West Indies (+2) → New Zealand (-2)
Hence, New Zealand Country has lost 4 matches most number of times.
34. (a)



So, women is related as wife of the Nirmal.

Solutions [Q. Nos. 35 to 37]

Condition I	Condition II	Condition III	Condition IV
6-C	6-	6... B	6... B
5-	5-C	5... D	5. B.D
4-A	4-	4-C	4. D
3... B	3-A	3-	3-C
2... B, D	2. B	2-A	2-
1-D	1... D	1-	1-A

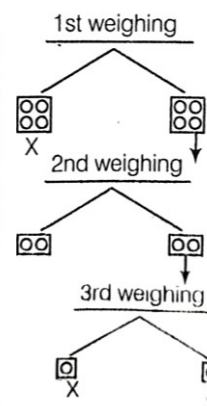
35. (d) According to condition IV, if A is standing on step 1, then D is standing one step higher than C.

36. (c) According to conditions I and II, if D is standing on step, 1 then A could be standing on either step 3 or 4 only.
37. (b) According to conditions I, III and IV, it there are two steps in between the steps A and D, then A may be standing in 4, 2 and 1 steps. So, in respect of option, we say that A must be standing on step 4.
38. (b) $A \times B = A = B$
 $A - B \Rightarrow A > B$
 $A + B \Rightarrow A < B$
 Then, Sachin * Mohan - Ravi
 $\Rightarrow$ Sachin = Mohan > Ravi
 Hence, Ravi is youngest among them.
39. (a) Total movement of hour hand = $20^\circ + 20^\circ = 40^\circ$
 So, hour hand covers $\frac{1^\circ}{2}$ in 1 min.
 Also, hour hand covers 40° in 80 min.
 Time duration to go tennis court and returning in (10+10) min

= 20 min.

∴ Total spent time at the tennis court = (80 - 20) min
 = 60 min

40. (c)



Hence, three times of weighing is required to identify the heavier ball.

